

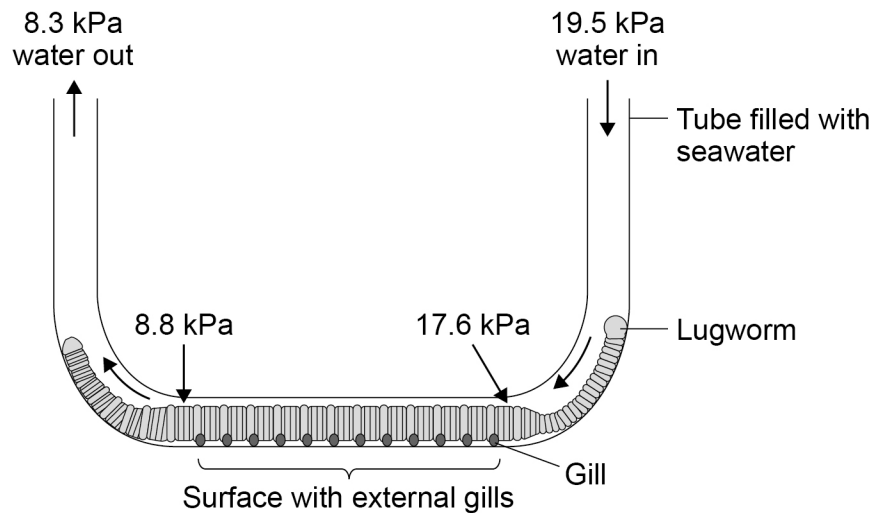
0 4 . 1

Lugworms create tubes in the sand on seashores. The tubes are filled with seawater.

A scientist measured the partial pressure of dissolved oxygen (pO_2) in seawater at different places in a tube with a lugworm inside.

Figure 3 shows her results.

Figure 3



The pO_2 of dissolved oxygen in lugworm blood is < 2.7 kPa

Using the data in **Figure 3**, what can you conclude about the uptake of oxygen over the entire body of the lugworm?

[4 marks]

Question 4 continues on the next page

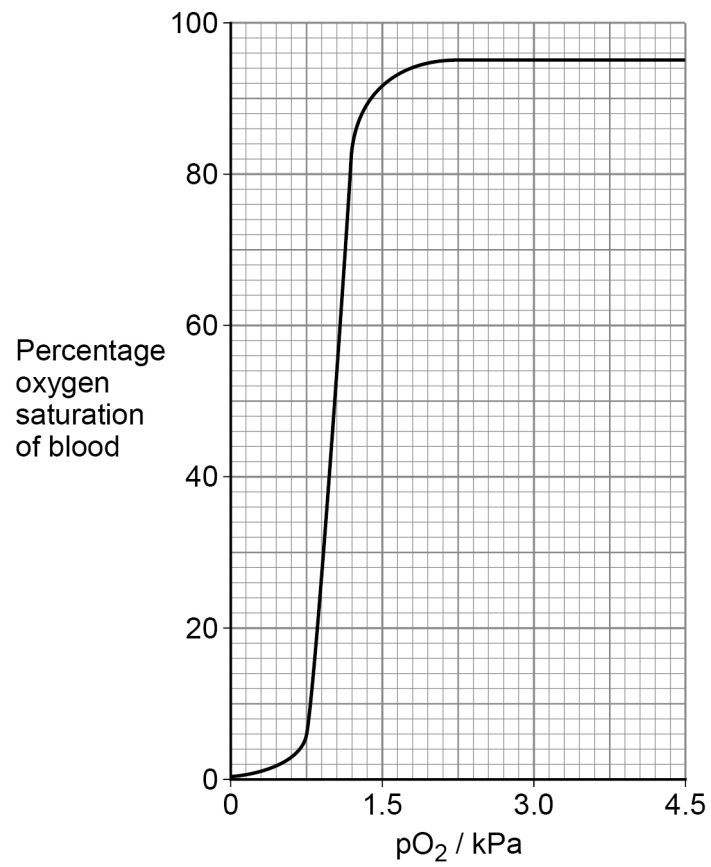
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0 4 . 2

Figure 4 shows the oxyhaemoglobin dissociation curve for a lugworm.

Figure 4



The oxygen saturation in the blood of a lugworm is 92%

The lugworm has 0.2 cm³ of blood.

Calculate the volume of dissolved oxygen in the blood of this lugworm using this equation

$$pO_2 = \frac{CdO_2}{0.000\ 031}$$

CdO₂ is the concentration of dissolved oxygen in the blood, with units cm³ oxygen per cm³ of blood.

Show your working.

[3 marks]

Answer _____ cm³

0 4 . 3

The intensity of the red colour in blood is affected by the pO₂ of the blood. The intensity of the colour in a solution is measured using a colorimeter.

The scientist used a colorimeter to measure the intensity of red colour in samples of lugworm blood with different pO₂ values. She prepared a calibration curve with this information.

Describe how the scientist will use information from the colorimeter and her calibration curve to determine the pO₂ in a sample of lugworm blood.

[2 marks]

